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Setup a standard edition standby in 1 hour

If you have Enterprise Edition, you have a great high availability feature that is Data Guard. According that you have setup cloud control as well, you have also a graphical console to manage your standby database. And if in addition to that you have diagnostic pack, then you can monitor it and send mail notification in case of apply gap. But EE licenses can be too high for a small business. In Standard Edition you can have a standby database as well, but then everything is manual: standby creation, log shipping, apply, failover. And doing a switchover is so complex that nobody does it manually.

In this article, I'll show how an automated setup can be easy, quick and reliable using Dbvisit Standby.

Dbvisit standby

Of course as a DBA, you like to do some scripting, in shell or perl, and can automate all the manual tasks that you need to manage a standby database. But that's not something easy to use, maintain and document. When it's about availability, we need something reliable. Dbvisit Standby is not a set of script made by the DBA. It's a real application maintained by developers, well documented and supported by a very responsive service desk. And we have an increasing set of features.

We have setup Dbvisit Standby at several customer sites, on different platforms and versions. Here I'll show how it is easy to setup with the GUI interface. The whole setup took 20 minutes. We have only to add the transfer time when building the standby database as it depends on the database size.

Here I'm replicating the SITE1 database which is on host VM211 and create a Dbvisit managed standby on VM212

ssh user equivalence

The first thing to do is to setup ssh user equivalence for both way passwordless ssh connections. That means:

- Generate a key with ssh-keygen
- Append the generated .ssh/id_rsa to .ssh/authorized_key in the other host
- Verify that you can ssh from one server to the other without a password:

```
[oracle@VM211 ~]$ ssh oracle@VM212
Last login: Tue Jul 1 11:29:45 2014 from 192.168.78.1
[oracle@VM212 ~]$ ssh oracle@VM211
Last login: Tue Jul 1 11:29:17 2014 from 192.168.78.1
[oracle@VM212 ~]$
```

Note that I did that with the version 6. The latest version do not need to rely on ssh and you can skip that step if you want.

Install the software

You need to create the destination directory:

```
[root@VM212 dbvisit]# mkdir /usr/local/dbvisit
[root@VM212 dbvisit]# chown oracle:dba /usr/local/dbvisit
```

Then unzip, untar and install dbvisit on both hosts:

```
[oracle@VM212 Dbvisit]$ tar -xf dbvisit-standby6.0.54.tar
[oracle@VM212 Dbvisit]$ cd dbvisit
[oracle@VM212 dbvisit]$ ls -l dbvisit_install
-rwxrwx---. 1 root vboxsf 8240115 Dec 15 2013 dbvisit_install
```

As you will see, dbvisit always provide a default value which is the right one, if you don't have a good reason for another value. I'll show only the relevant entries:

```
[oracle@VM212 dbvisit]$ ./dbvisit_install
...
> Oracle user name on this server? [default: "oracle"]
...
> Dbvisit installation directory path? [default: "/usr/local/dbvisit"]
...
> Dbvserver admin user name? [default: "admin"]
...
> Web interface? [default: "https"]
...
> Listening port number? [default: "8443"]
...
> Turn on automatic email to Dbvisit support: Yes/No [default: "Yes"]
...
> Start Dbvserver: Yes/No [default: "Yes"]
```

Then you just have to read the next steps which are displayed at the end of the install log.

It's good to have dbvisit started at server reboot, so let's setup the init script:

```
[root@VM211 init.d]# cp /usr/local/dbvisit/dbvserver/etc/init.d/redhat/
dbvserverd /etc/rc.d/init.d/dbvserverd
[root@VM211 init.d]# chkconfig --add dbvserverd
```

All that is done on both servers.

Once done, you don't have to go to the standby server anymore.

Now, this is all we have to do on command line, the server is started and we have a url to connect to:

```
i) Please point your web browser to the following URL to login to
Dbvserver and configure your Dbvisit product:
https://192.168.78.211:8443
Login user name is 'admin' (password 'admin').
```

The console will show the following menu:

- Setup: create standby, update configuration, synchronize etc.
- Run: run interactive commands (start databases, switchover, open read only) or set schedules
- Reporting: graphical history of transfer time, gaps, etc.
- Logs: quickly check Dbvisit log or database alert.log (good alternative to tail-f)



Graphical Setup

You don't need a documentation for that. Once connected to the console just click on Setup and you will define a few configuration parameters, most of them having a relevant default value.

You can reconfigure an existing standby database but here we are creating a new setup:



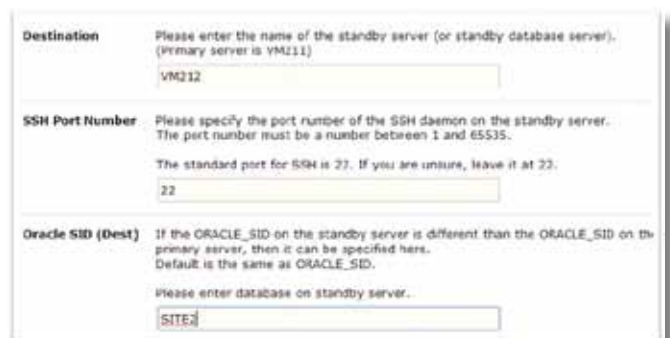
You select the primary database for which you want to create a standby, it's just a dropdown box with all the instances you have on the server.



And then you navigate upon the steps to create the configuration:



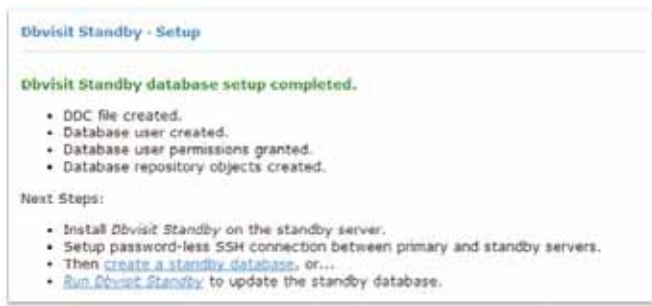
I'll not show all the screens here. Each option has a description and a default value which fit most of the cases:



I really like the 'if you are unsure, live it at the default value' message. With description, help box and default value, you can't hesitate.

Create the standby

Once the setup is completed we have now to create the standby database. No need to do it manually. Of course you can do it yourself if you prefer, with RMAN DUPLICATE for example. But we will just continue on the friendly screens:



And again we have a few parameters to review and change if needed.

From the beginning (software installation) up to this point, it took only 20 minutes. 5 minutes to setup user equivalence, 5 minutes to install the software and 10 minutes to configure. You can plan 15 additional minutes if you want to read all parameter descriptions quickly. Even for my first Dbvisit Standby installation, I haven't read the manual: the wizard is straightforward.

Now the time to create the database depends on the size. We have an estimation, which is very low here as I'm doing it on an empty database:



Depending on the size of your database and the network bandwidth between both hosts, you have the possibility to:

- Compress files for the transfer
- Write files to a media that you will ship yourself (external disk)

Then everything run automatically:



And our standby database is ready on host VM212.



113 Sicherheitsupdates für die Oracle-Produktpalette

Oracle hat am 14. Juli sein vierteljährliches Critical Patch Update (CPU) veröffentlicht. Insgesamt beseitigt der Software-Hersteller 113 Sicherheitslücken in einer Vielzahl von Produktfamilien.

Neben den Datenbanken Oracle und MySQL sind auch die Fusion Middleware, Enterprise Manager Grid Control, Java SE, die Virtualisierungstechnologien sowie die Betriebssysteme Linux und Solaris von dem Patch betroffen. Darüber hinaus schließt Oracle auch viele Lecks in den Business Applications.

Besonders betroffen vom CPU ist Java SE mit 20 Updates. Eine Schwachstelle erreicht den höchsten CVSS-Base-Score von 10.0, während sieben weitere Lecks in der Java Hotspot Virtual Maschine, Java FX und in den Libraries mit einem Wert von 9,3 abschließen. Alle 20 Lücken ermöglichen ein unbefugtes Eindringen über das Netzwerk.

Die Oracle Fusion Middleware erhält 29 neue Sicherheitsupdates. Der schwerwiegendste CVSS Base-Score für diese Schwachstellen ist 7.5. Neben Weblogic, GlassFish, JDeveloper und Webcenter Portal sind auch Oracle Traffic Director und iPlanet WebProxy Server sowie Oracle http Server im Visier der Aktualisierungen.

Fünf Schwachstellen in der Oracle-Datenbank

Darüber hinaus werden fünf Schwachstellen in der Oracle-Datenbank geschlossen. Keine von ihnen ermöglicht einen unbefugten Zugriff über das Netzwerk. Der höchste CVSS-Score beträgt 9.0.

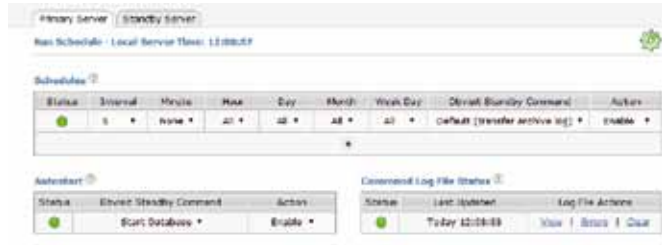
Für MySQL stellt Oracle weitere zehn Updates zur Verfügung, während 15 Aktualisierungen im Virtualisierungsumfeld vor allem Oracle Secure Global Desktop und die Virtual Box zugutekommen. In Solaris werden vier Lecks beseitigt.

Ferner schließt Oracle im Applicationsumfeld fünf Schwachstellen in Hyperion und fünf weitere für die E-Business-Suite. Es kommen weitere fünf Aktualisierungen für PeopleSoft und drei für die Supply Chain Suite. Neben den sechs Patches in Siebel werden auch Lecks in den Oracle Industry Applications behandelt.



Schedule log shipping

Time to ship the archived logs and apply them. We need to run dbvisit on both servers. When it's in the primary site, it will archive the current log file and ship it to the standby site. When it's in the standby site, it does the apply. We can add an entry in the crontab for that, or use the scheduler that comes with Dbvisit. Here I define it with the console to be run by the scheduler every 5 minutes:



And the same for the standby server.



How did I choose the interval? In case of primary node failure, I want to minimize the lost transactions. Here I'm sure that the gap will always be lower than 5 minutes. What's the cost to switch log every 5 minutes? Except if you have a huge number of datafiles to update at checkpoint, the overhead of is negligible.

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Note that when you can setup from crontab, then it's a good idea to have a small delay so that the primary log has the time to be shipped to the standby server, for example:

VM211:

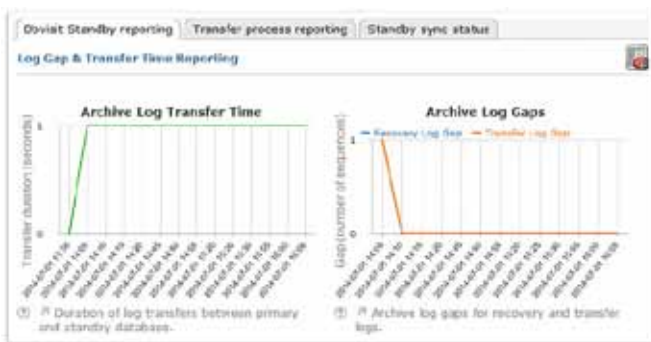
```
00,05,10,15,20,25,30,35,40,45,50,55 * * * * /usr/local/dbvisit/standby/dbvisit SITE1
```

VM212:

```
02,07,12,17,22,27,32,37,42,47,52,57 * * * * /usr/local/dbvisit/standby/dbvisit SITE1
```

Reporting

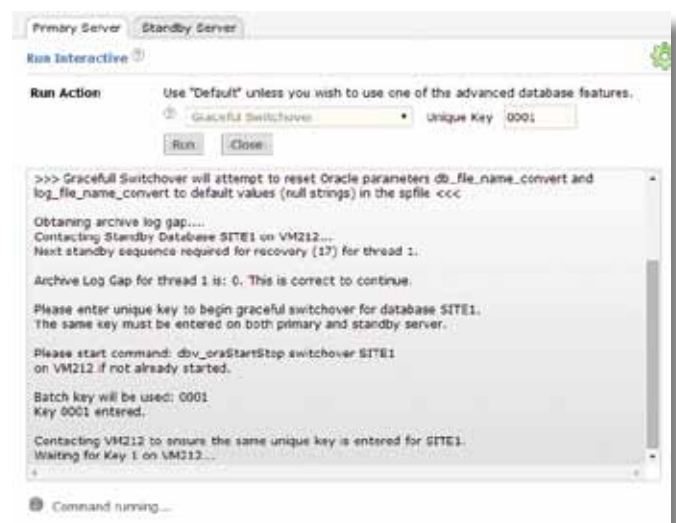
Besides the monitoring that can be setup to send a mail in case of error, the main indicators that we check are the transfer time and the log gap. Here we see that as soon as I've activated the scheduler jobs, I've no gap.



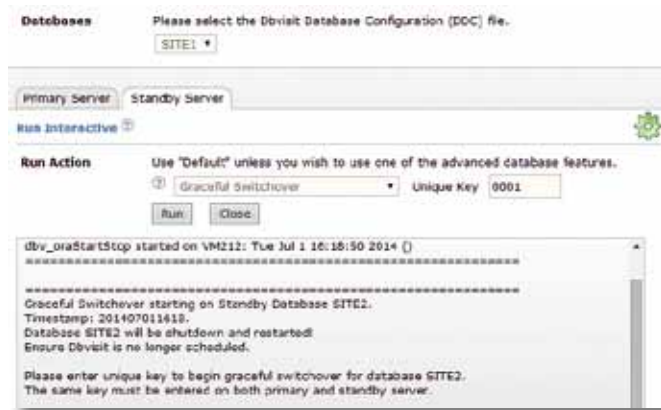
About the notifications, you can configure to send an e-mail only in case of failure or gap. But you can choose to send success mail as well so that you are sure that the notification is working well.

Switchover

There is one remaining task to do: test our standby configuration. I'll do a switchover: It is just two clicks: we initiate the switchover on both servers, giving a key so that we are sure that it's the same operation. It's graphical and we can follow the steps from the console.



It waits for the switchover to be initiated on the standby as well:



This is fully automated. It's not as fast as DataGuard but you can still switchover in a few minutes. The point is reliability: the steps are done one after the other to be sure that the config is not left in an intermediary state.

Remember, as with Data Guard, a physical standby is there to be used. Do regular switchovers, not only when you have a maintenance on a server, but also at regular intervals during a planned outage. It's the key for no stress high availability: because you do it frequently, you are ready to switchover or failover without any fear when an event occurs.

Of course, the client must follow the primary database. Network setup is the same as with Data Guard: either have a single place to update a tnsnames.ora or use Transparent Application Failover.

Conclusion

Dbvisit standby is a very nice solution to bring high availability to Standard Edition without the additional complexity to maintain DBA-made scripts. It is a robust software that keep it simple and reliable. This demo setup actually took me less than one hour and anybody can do it by downloading the trial version from <http://www.dbvisit.com/products/downloads>. Play with it. With an affordable cost you can have the safety of a standby database ready to recover from any failure – without having to setup a new server and/or restore a backup.

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Oracle ADF Mobile heisst jetzt Oracle MAF

Bisher hiess das Model-View-Controller-Framework ADF Mobile. Seit Ende Juni nennt sich das Werkzeug zur Entwicklung von mobilen Multi-Channel-Applikationen nun Oracle Mobile Application Framework (Oracle MAF). Verfügbar ist es nicht nur als Extension vom JDeveloper, sondern jetzt auch als Eclipse-Extension (über Oracle Enterprise Pack for Eclipse). Aufgrund der Nähe zu ADF Mobile dürfte die Migration zu MAF denkbar trivial sein. Die «ADF Mobile»-Applikation müsse nur im JDeveloper 12.1.3 aufgemacht werden, erklärt der Oracle-Produktverantwortliche Shay Shmeltzer in seinem Blog.

Neben neuen UI-Komponenten und der Integration der Cordova-Plugins hat sich mit dem neuen Release vor allem in Sachen Lizenzierung was getan: Oracle MAF kann als eigenständiges Produkt lizenziert werden und verlangt demnach nicht unbedingt eine Weblogic- oder ADF-Lizenz. Das Produkt kann darüber hinaus entweder auf Basis der Benutzer (Named User Plus) oder auf Basis der entwickelten Applikationen (Application Developed) lizenziert werden.

